



Non-Verbal Reasoning



The non-verbal reasoning test involves solving problems that are represented with pictures and diagrams. There are a number of different types of questions for the non-verbal reasoning test which could include:

- finding shapes that are like each other based on similarities (for example colours and number of sides);
- finding the odd one out in a group of pictures;
- identifying a shape which has been rotated;
- breaking the code in a sequence of images;
- spotting the missing shape in a sequence;
- identifying the nets of 3D shapes.



Here is an example of the type of question you may get in this section:



1. Which is the correct code for figure 4?

 M B	 R B	 M Q	 ?
1	2	3	4

B	Q	B	R
Y	M	Q	Q
A	B	C	D

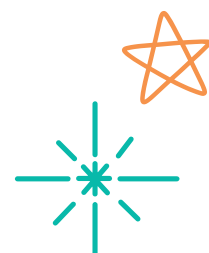


The answer would be D. The letter common to the first two shapes is B which represents the cross-hatching pattern they both have. Therefore Q represents the solid orange colour. The M is the letter representing the shape of the moon and it comes at the top of the box. Therefore the answer is D (R to represent the irregular dodecagon and the Q at the bottom to represent the solid orange colour).

The non-verbal reasoning test, for many children, can be very intimidating. The best way to prepare for this is through practice. Complete many different type of practice non-verbal reasoning tests with your child and discuss how each question is solved. This will give your child the experience needed to tackle the on-verbal reasoning tests.



For a thorough guide to all aspects of the 11+ test, try our [Survival Guide for Parents/Carers](#).



11+ Non-Verbal Reasoning Practice Paper

Analogies

An **analogy** is when we compare two things based on a **relationship** they share.

For example, look at the statement below:

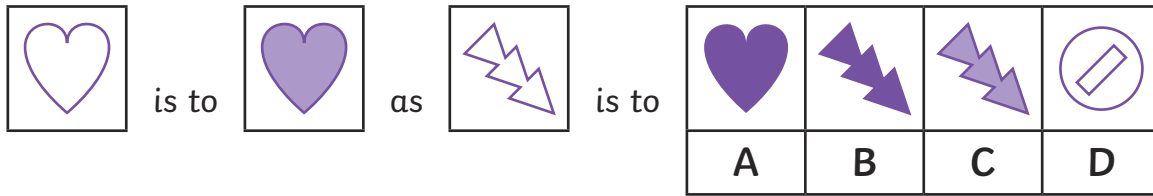
Cold is to hot as old is to young.

The relationship between **cold** and **hot** and **old** and **young** is that they are opposites.

In this quiz, you must work out the **analogy** between the first two images. This will help you to find an analogy between the third picture and one of the following four choices.

For example:

Which one of the figures in the boxes completes the analogy?



The answer is **c**.

In the example, the first shape is a heart and so is the second shape. This tells us that the figure we are looking for needs to be the same shape as the third figure, so we are looking for a lightning bolt shape. The first shape has changed from white to lightly coloured, so we are looking for a lightly coloured shape. The lightly coloured lightning bolt is figure **c**.

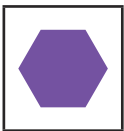


11+ Non-Verbal Reasoning Practice Paper

Analogies: Test 1

Which one of the figures in the boxes completes the analogy?

1.

 is to  as  is to

			
A	B	C	D

 answer

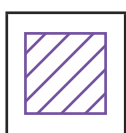
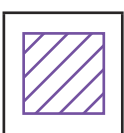
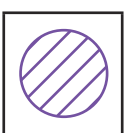
2.

 is to  as  is to

			
A	B	C	D

 answer

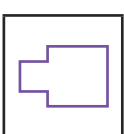
3.

 is to  as  is to

			
A	B	C	D

 answer

4.

 is to  as  is to

			
A	B	C	D

 answer

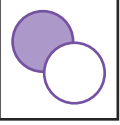
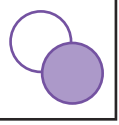
5.

 is to
  as
  is to

			
A	B	C	D

answer

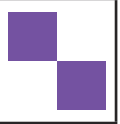
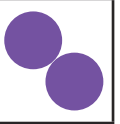
6.

 is to
  as
  is to

			
A	B	C	D

answer

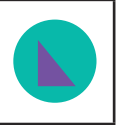
7.

 is to
  as
  is to

			
A	B	C	D

answer

8.

 is to
  as
  is to

			
A	B	C	D

answer



11+ Non-Verbal Reasoning Practice Paper:

Analogies: Test 1 Answers

1.

 is to
  as
  is to

			
A	B	C	D

answer B

The answer is B because the relationship between the first image and the second image is that the colour changes from orange to purple.

2.

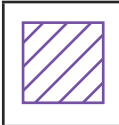
 is to
  as
  is to

			
A	B	C	D

answer C

The answer is C because the relationship between the first image and the second image is that the image is rotated 180°.

3.

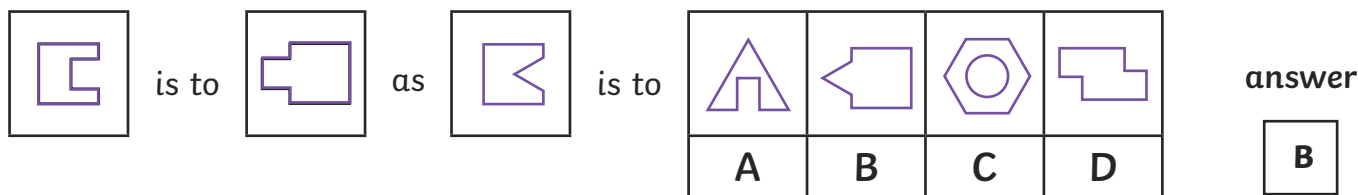
 is to
  as
  is to

			
A	B	C	D

answer D

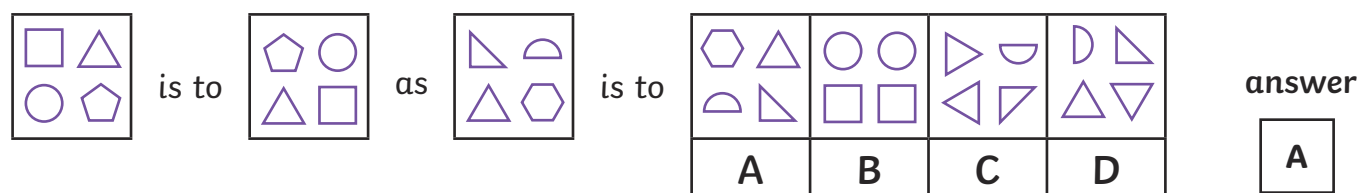
The answer is D because the relationship between the first image and the second image is that the shape stays the same, but the direction of the diagonal lines within the shape changes.

4.



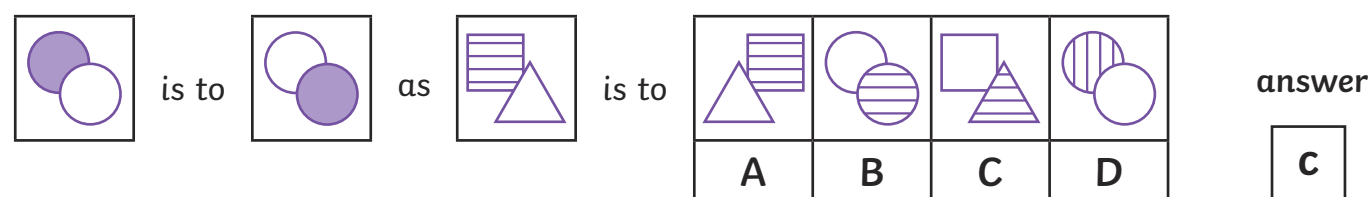
The answer is B because the relationship between the first image and the second image is that the shape missing from the first figure has been added to the side of the second figure.

5.



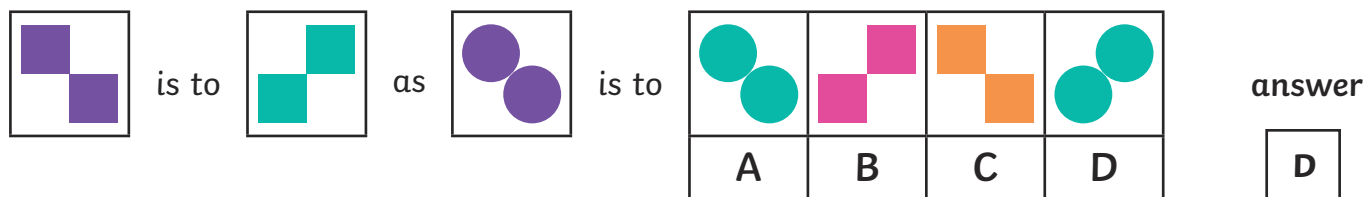
The answer is A because the relationship between the first image and the second image is that the shapes are in the opposite position. For example, the square moves from top left to bottom right and the circle moves from bottom left to top right.

6.



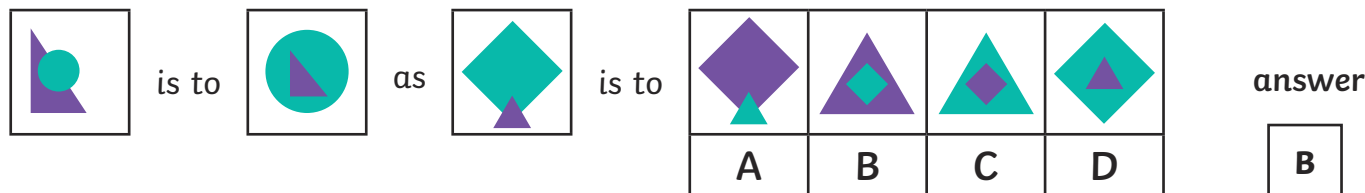
The answer is C because the relationship between the first image and the second image is that the shapes swap colours. The third image has patterns rather than colour, so we are looking for a figure that has the shapes in the same position and order – a square behind and a triangle in front – but the square should be blank and the triangle should now be patterned.

7.



The answer is D because the relationship between the first image and the second image is that the figure is mirrored and the colours change from purple to teal.

8.



The answer is B because the relationship between the first image and the second image is that the smaller shape becomes larger and the larger shape goes inside it. So we are looking for a large dark triangle with a small light square inside.

11+ Non-Verbal Reasoning Practice Paper

Analogies: Test 2

Which one of the figures in the boxes completes each analogy?

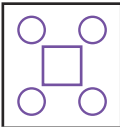
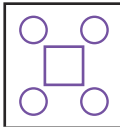
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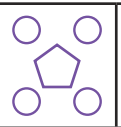
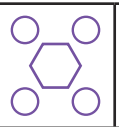
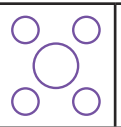
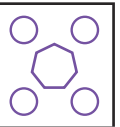
 is to  as  is to

			
A	B	C	D

answer

2.

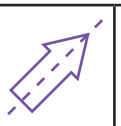
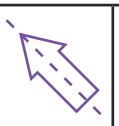
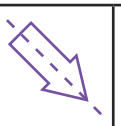
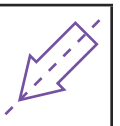
 is to  as  is to

			
A	B	C	D

answer

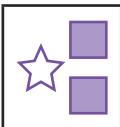
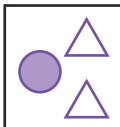
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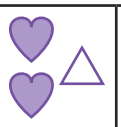
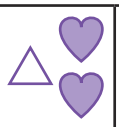
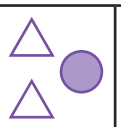
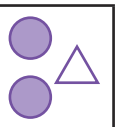
 is to  as  is to

			
A	B	C	D

answer

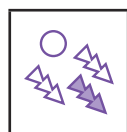
4.

 is to  as  is to

			
A	B	C	D

answer

5.



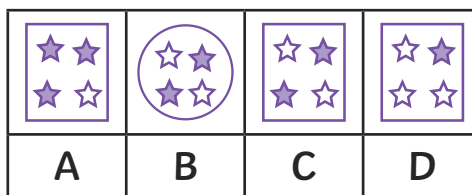
is to



as



is to

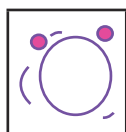


answer

6.



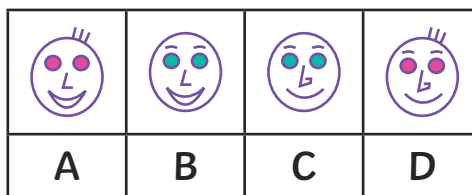
is to



as



is to



answer

7.



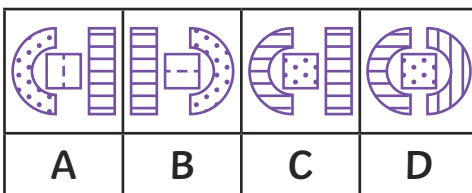
is to



as

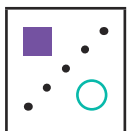


is to

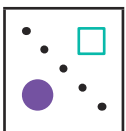


answer

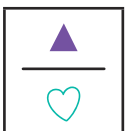
8.



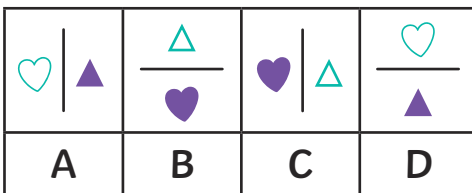
is to



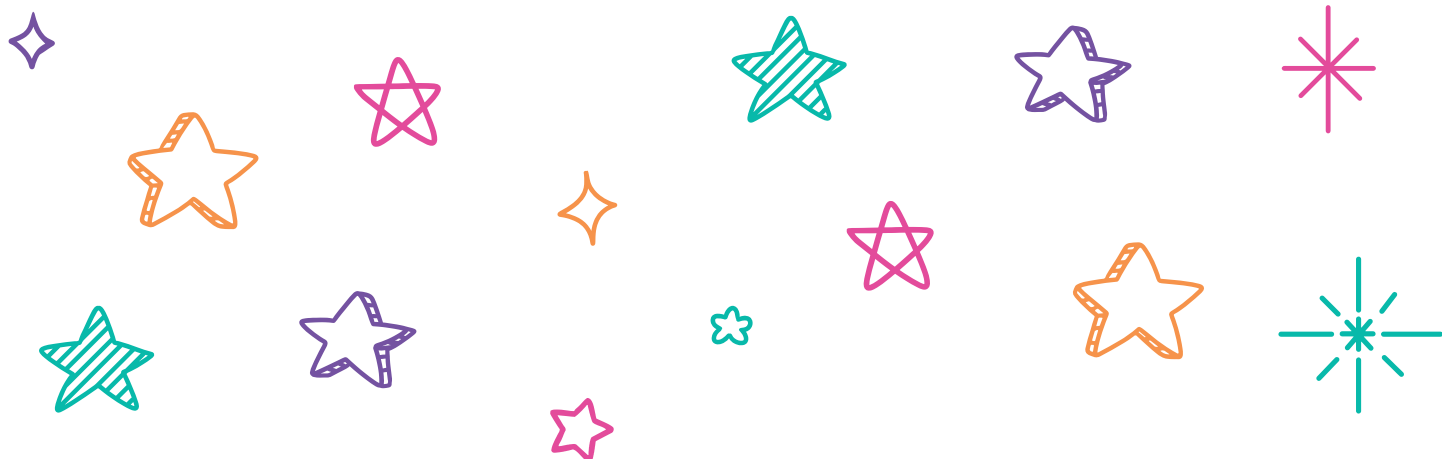
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is to



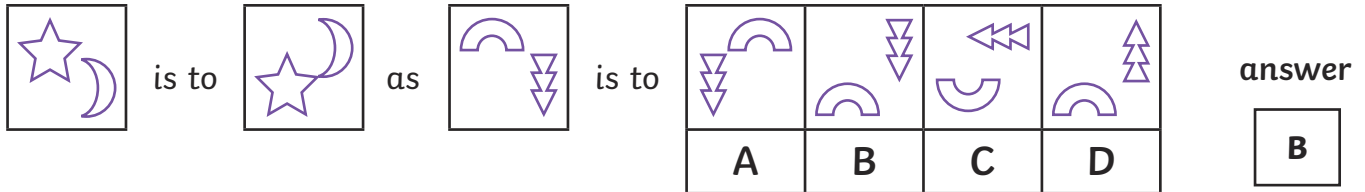
answer



11+ Non-Verbal Reasoning Practice Paper:

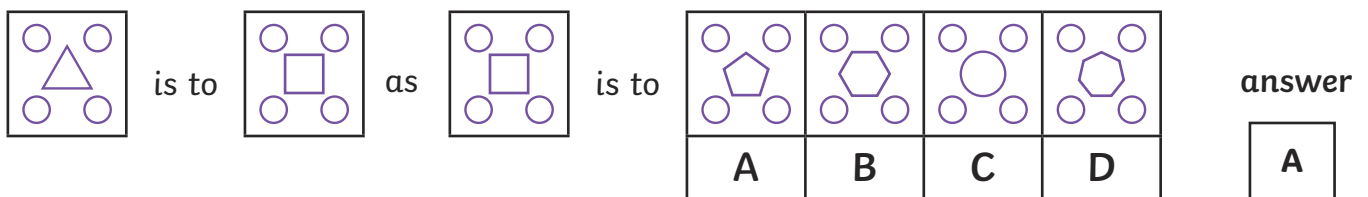
Analogies: Test 2 Answers

1.



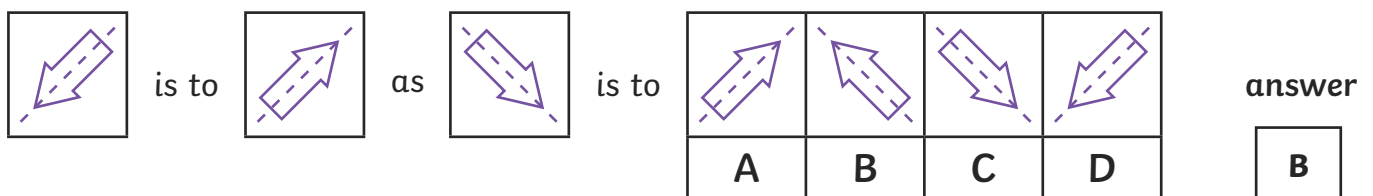
The answer is B because the relationship between the first image and the second image is that the star shape has moved down and the moon shape has moved up. This means that the curved shape needs to move down and the arrow shape needs to move up.

2.



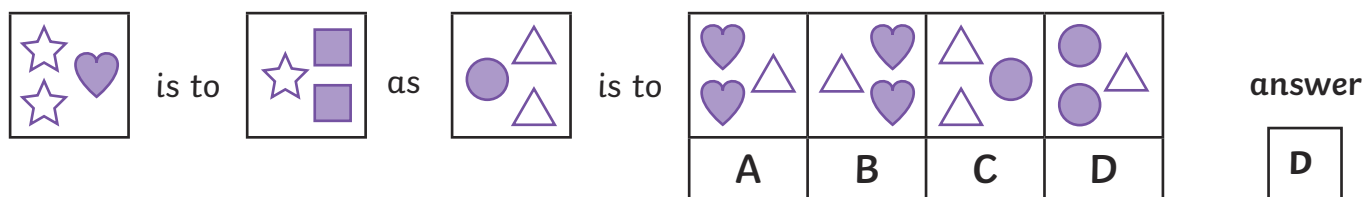
The answer is A because the relationship between the first image and the second image is that the outer shapes stay the same, but the inner shape has changed to a shape with one more side. That means that we are looking for a shape with one more side than a square for the fourth figure, which is a pentagon as shown in A.

3.



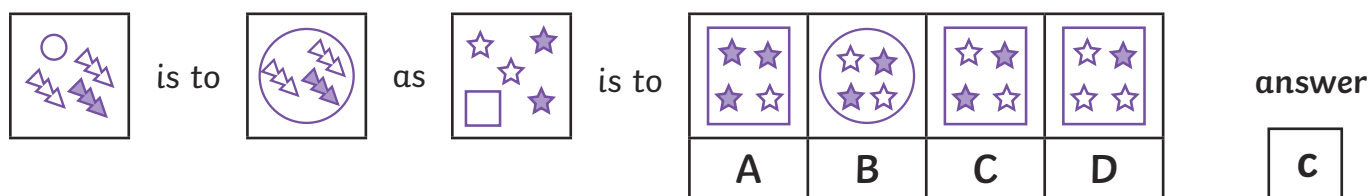
The answer is B because the relationship between the first image and the second image is that the arrow changes to face the opposite direction.

4.



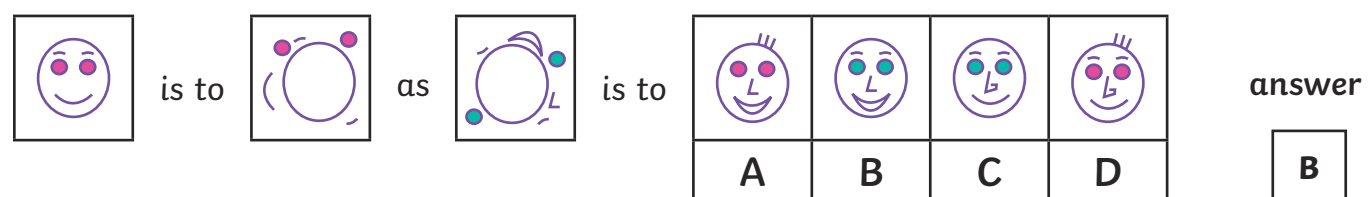
The answer is D because the relationship between the first and second image is that the number of shapes on the left is halved and the number of shapes on the right is doubled. We are therefore looking for one purple circle on the left and two white triangles on the right.

5.



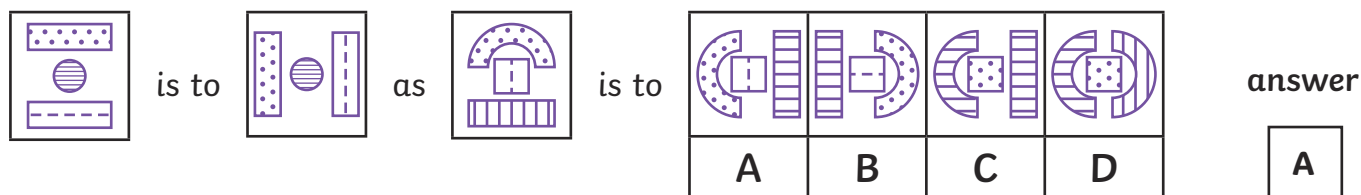
The answer is C because the relationship between the first image and the second image is that the odd shape out becomes larger, with the other shapes going inside it. So we are looking for a square with two white stars and two dark stars inside it.

6.



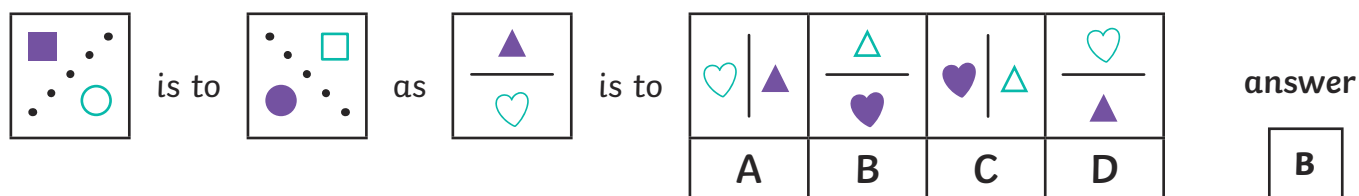
The answer is B because the relationship between the first image and the second image is that the facial features have been removed from the face and placed outside. We can tell by the nose and mouth shapes as well as the eye colours that figure 3 can be put together to make option B.

7.



The answer is A because the relationship between the first image and the second image is that the outer shapes move 90° anticlockwise around the middle shape, which stays exactly the same. A is the only option in which the shapes have been moved in this way.

8.



The answer is B because the relationship between the first image and the second image is that the figure is mirrored and the shapes swap colours.